Lecture 3 v2:
Simple Linear Regression

25117 - Econometrics

Universitat Pompeu Fabra

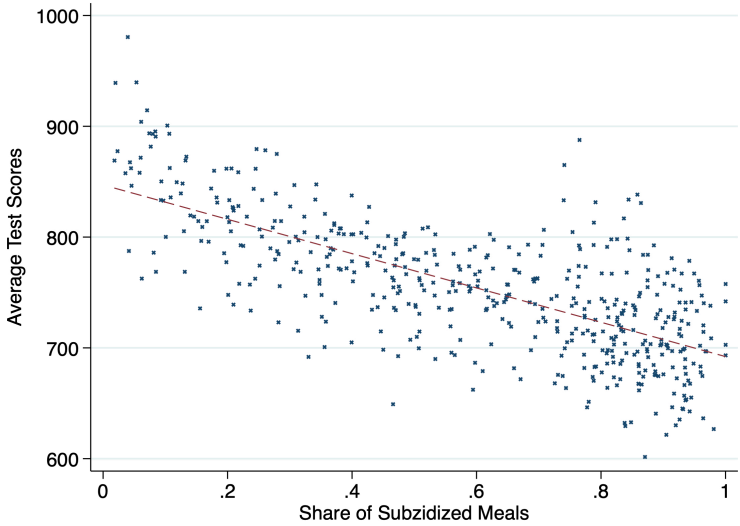
October 7th, 2024

What we learned in the last lesson

- The sample average, $\overline{Y}$, is an estimator of the population mean, μ_Y. When $Y_1, \ldots, Y_n$ are i.i.d.,
  1. The sampling distribution of $\overline{Y}$ has mean μ_Y and variance $\sigma^2_{\overline{Y}} = \sigma^2_Y/n$
  2. $\overline{Y}$ is unbiased
  3. by the LLN, $\overline{Y}$ is consistent; and
  4. by the CLT, $\overline{Y}$ has an approximately normal sampling distribution when the n is large.
- The t-statistic is used to test the null hypothesis that the population mean takes on a particular value. If n is large, the t-stat $\sim \mathcal{N}(0,1)$ under H_0.
- The t-statistic can be used to calculate the p-value associated with the null hypothesis. The p-value is the probability of drawing a statistic at least as adverse to the null hypothesis as the one you actually computed in your sample, assuming the null hypothesis is correct. A small p-value is evidence that the null hypothesis is false.
- A 95% confidence interval for μ_Y contains the true value of μ_Y in 95% of all possible samples.
- The sample correlation coefficient is an estimator of the population correlation coefficient and measures the linear relationship between two variables – that is, how well their scatterplot is approximated by a straight line.

The regression line

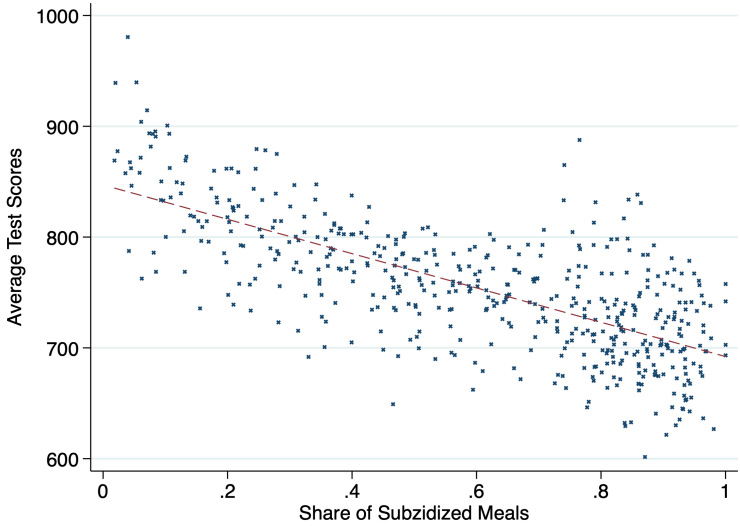
- The population regression line is the expected value of Y given X.
- The slope is the difference in the expected values of Y, for two values of X that differ by one unit



The regression line

The estimated regression can be used either for

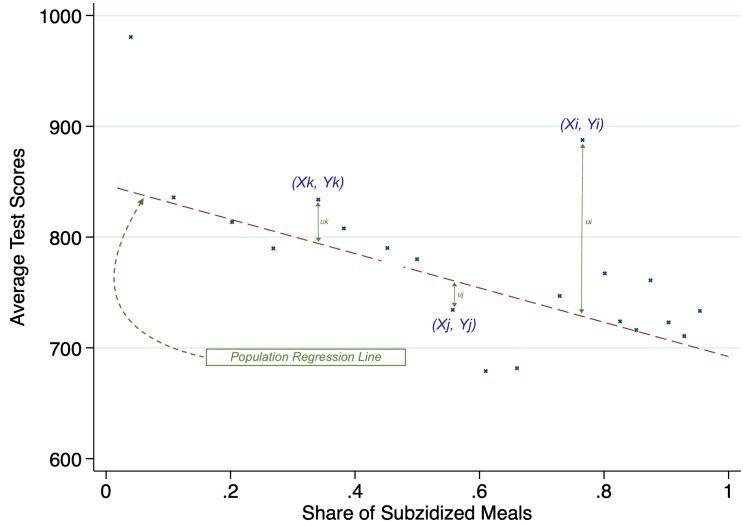
1. **causal inference** (learning about the causal effect on Y of a change in X)
   - → What is the causal impact of subsidizing meals on students' performances?
2. **prediction** (predicting the value of Y given X, for an observation not in the data set)
   - → What are the students' expected test scores given the share of subsidized meals?



The Population Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- There are n observations, (X_i, Y_i), $i = \{1, \ldots, n\}$
- Y is the dependent variable
- X is the independent variable or regressor
- $\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 X$ is the population regression line
- β_0 is called the intercept
- β_1 is the slope
- u_i is the error term



The Ordinary Least Squares (OLS) Estimator

How can we estimate β_0 and β_1 from the data? Recall from Lecture 1 that the Least Square estimator of μ_Y, $\overline{Y}$, solves

$$\min_m \sum_{i=1}^{n} (Y_i - m)^2$$

By analogy, the OLS estimators of the unknown parameters β_0 and β_1, $\hat{\beta}_0$ and $\hat{\beta}_1$ solves,

$$\begin{aligned} & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} u_i^2 \\ \Leftrightarrow \quad & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - \mathbb{E}(Y \mid X))^2 \\ \Leftrightarrow \quad & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 \end{aligned}$$

The Ordinary Least Squares (OLS) Estimator

$$
\begin{aligned}
& \begin{cases} \frac{\partial}{\partial \beta_0} \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 = 0 \\ \frac{\partial}{\partial \beta_1} \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 = 0 \end{cases} \\
\Rightarrow \quad & \begin{cases} -2 \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)) = 0 \\ -2 \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)) X_i = 0 \end{cases} \\
\Rightarrow \quad & \begin{cases} \frac{1}{n} \sum_{i=1}^{n} Y_i - \frac{1}{n} \sum_{i=1}^{n} \hat{\beta}_0 - \frac{1}{n} \sum_{i=1}^{n} \hat{\beta}_1 X_i = 0 \\ \frac{1}{n} \sum_{i=1}^{n} Y_i X_i - \frac{1}{n} \sum_{i=1}^{n} \hat{\beta}_0 X_i - \frac{1}{n} \sum_{i=1}^{n} \hat{\beta}_1 X_i^2 = 0 \end{cases} \\
\Rightarrow \quad & \begin{cases} \overline{Y} - \hat{\beta}_0 - \hat{\beta}_1 \overline{X} = 0 \\ \frac{1}{n} \sum_{i=1}^{n} Y_i X_i - \hat{\beta}_0 \overline{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^{n} X_i^2 = 0 \end{cases}
\end{aligned}
$$

The Ordinary Least Squares (OLS) Estimator

$$
\begin{aligned}
&\Rightarrow \quad \begin{cases} \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} \\ \frac{1}{n}\sum_{i=1}^{n} Y_i X_i - \overline{YX} - \hat{\beta}_1 (\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{XX}) = 0 \end{cases} \\
&\Rightarrow \quad \begin{cases} \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} \\ \hat{\beta}_1 = \frac{\frac{1}{n}\sum_{i=1}^{n} Y_i X_i - \overline{YX}}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{XX}} \end{cases} \\
&\Rightarrow \quad \begin{cases} \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} \\ \hat{\beta}_1 = \frac{\frac{1}{n}\sum_{i=1}^{n} (Y_i - \overline{Y})(X_i - \overline{X})}{\frac{1}{n}\sum_{i=1}^{n} (X_i - \overline{X})^2} \end{cases} \\
&\Rightarrow \quad \begin{cases} \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} \\ \hat{\beta}_1 = \frac{s_{XY}}{s_X^2} \end{cases}
\end{aligned}
$$

Back to the Linear Regression Model

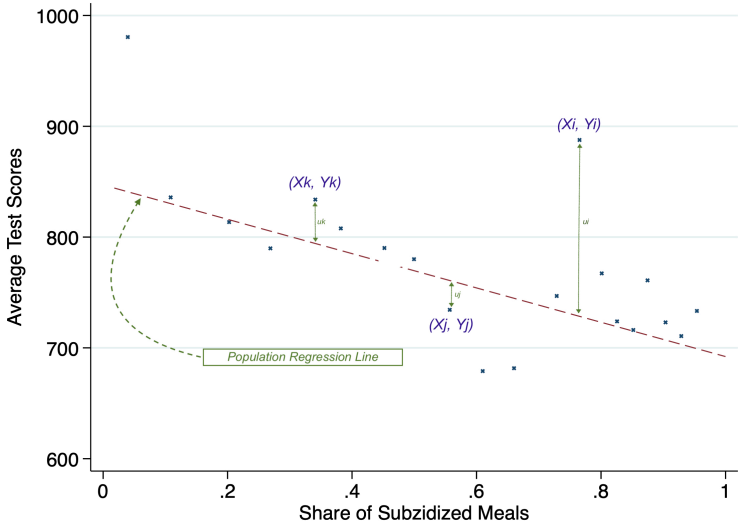
The OLS estimator minimizes the average squared difference between the actual values of i Y and the prediction ("predicted value") based on the estimated line.

- $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}$
- $\hat{\beta}_1 = \frac{s_{XY}}{s_X^2}$
- $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$
- $\hat{u}_i = Y_i - \hat{Y}_i$

This means that

- $\Delta \mathbb{E}(Y_i \mid X_i) = \hat{\beta}_1 \Delta X_i$

s



Interpretation

Here, the population regression line is given by

$$\hat{\beta}_0 \approx 754 + 155 \times .6 \approx 847$$
$$\hat{\beta}_1 \approx \frac{-11.78}{.28^2} \approx -155$$
$$\hat{Y}_i \approx 847 - 155 \times X_i$$

- This means that a 10ppt increase in the share of subsidized meals is associated with ($\neq$ cause!) a 15.5 decrease in the average test scores.
- To give a standard of comparison, this is about 26% of the Test scores' standard deviation!
- The intercept (taken literally) means that, according to this estimated line, districts with zero meals would have a (predicted) test score of approximately 847...
- But this interpretation of the intercept makes no sense – it extrapolates the line outside the range of the data (see Lect2 for descriptive stats) – here, the intercept is not economically meaningful.

Goodness of Fit

- How well does our regression model fit the data? This is a crucial question.
- Goodness of fit measures help us assess the quality of our model.
- Two regression statistics provide complementary measures of how well the regression line "fits" or explains the data:
  1. The regression R^2 measures the fraction of the variance of Y that is explained by X; it is unitless and ranges between zero (no fit) and one (perfect fit)
  2. The standard error of the regression (SER) measures the magnitude of a typical regression residual in the units of Y.
- These measures are very useful to predict Y when X is known but Y is not (out-of-sample prediction). A common prediction measure is $\hat{Y} \pm SER$.

R^2

From the previous slides, we note that

$$Y_i = \hat{Y}_i + \hat{u}_i$$

The R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y

$$\begin{aligned} R^2 &= \frac{\text{explained sum of squares } (ESS)}{\text{total sum of squares } (TSS)} = \frac{\sum_{i=1}^{n}(\hat{Y}_i - \overline{Y})^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} \\ &= 1 - \frac{\text{sum of squared residuals } (SSR)}{\text{total sum of squares } (TSS)} = 1 - \frac{\sum_{i=1}^{n}\hat{u}_i^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} \end{aligned}$$

R^2 ranges from 0 to 1, where higher values indicate a better fit (i.e., the regressor is good at predicting the outcome)

The Standard Error of the Regression (*SER*)

The SER measures the spread of the distribution of u around the regression line in the units of the dependent variable. It is an estimator of the standard deviation of the regression error u_i.

$$SER = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(\hat{u}_i - \overline{\hat{u}}_i)^2} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2} = \sqrt{\frac{SSR}{n-2}}$$

Here, division by $n-2$ is a dof correction (two parameters, β_0 and β_1 have been estimated, whereas in s_Y^2 only one has been estimated: μ_Y, with $\overline{Y}$)

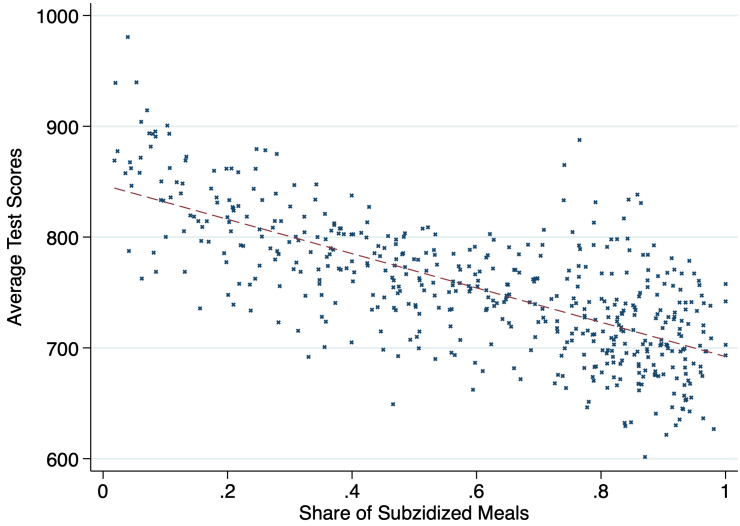
The root mean squared error (*RMSE*) is closely related to the *SER*:

$$RMSE = \sqrt{\frac{SSR}{n}}$$

In practice, that does not change anything for large samples. Both are interesting as they provide a consistent estimate of the standard deviation in the unobservables affecting Y.

Goodness of Fit

- Here the $R^2 = .509$ and $RMSE = 42.601$
- About half of the variation in actual test scores is explained by the variation in the predicted test scores (which, in our regression specification, comes from the variation in subsidized meals).
- Does this make sense? Does this mean that subsidizing school meals matters for education in a policy sense?



The conditions for causality

The OLS Assumptions for Causal Inference

Let

$$Y_i = \beta_0 + \beta_1 X_i + u_i \;\; ; \;\; i = 1, ..., n$$

where β_1 is the causal effect of X on Y. Then:

1. The error term u_i has conditional mean 0 given X_i: $\mathbb{E}(u_i \mid X_i) = 0$;
2. (X_i, Y_i) $i = 1, ..., n$, are independent and identically distributed (i.i.d.) draws from their joint distribution; and
3. Large outliers are unlikely: X_i and Y_i have nonzero finite fourth moments.

The conditions for causality

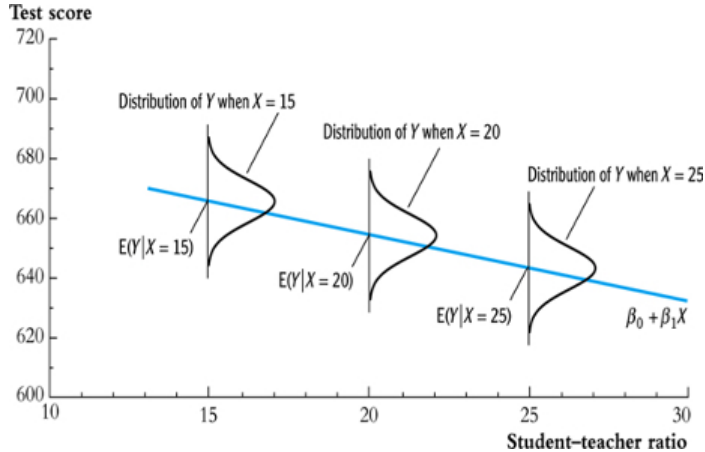
If these assumptions hold, then

1. In large samples, the OLS estimators are consistent and have sampling distributions that are normal allowing for testing hypotheses and constructing confidence intervals
2. They organize the circumstances that pose difficulties for OLS estimation of the causal effects
   - Assumption (1) ensures that $\hat{\beta}_1$ is an unbiased estimator of β_1
   - Assumption (2) holds in cross-sectional contexts but is inappropriate for panel and time series data
   - Assumption (3) reminds the importance of checking for large outliers

Do you think these assumptions hold in our California Schools Experiment?

Assumption (1) : $\mathbb{E}(u_i \mid X_i) = 0$

- If $E(u_i|X_i) \neq 0$, it implies that there is systematic variation in Y not explained by X.
- This unexplained variation can confound our causal interpretations, making it challenging to isolate the true causal effect of X on Y.



Assumption (1) : $\mathbb{E}(u_i \mid X_i) = 0$

Remember that

$$\hat{\beta}_1 = \frac{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \overline{Y})(X_i - \overline{X})}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2} = \frac{s_{XY}}{s_x^2} = \frac{cov(X, Y)}{var(X)}$$

Then

$$\begin{aligned}
\hat{\beta}_1 &= \frac{cov(X, [\beta_0 + \beta_1 X + u])}{var(X)} \\
&= \frac{cov(X, \beta_0)}{var(X)} + \frac{cov(X, \beta_1 X)}{var(X)} + \frac{cov(X, u)}{var(X)} \\
&= \beta_1 + \frac{cov(X, u)}{var(X)}
\end{aligned}$$

If $\mathbb{E}(u_i \mid X_i) \neq 0$, then $cov(X, u) \neq 0$... In the California Schools Experiment, schools subsidizing meals (vs. not) are likely to welcome students from poor (vs. rich) economic and social backgrounds – biasing estimates downwards!

Assumption (2): i.i.d. draws

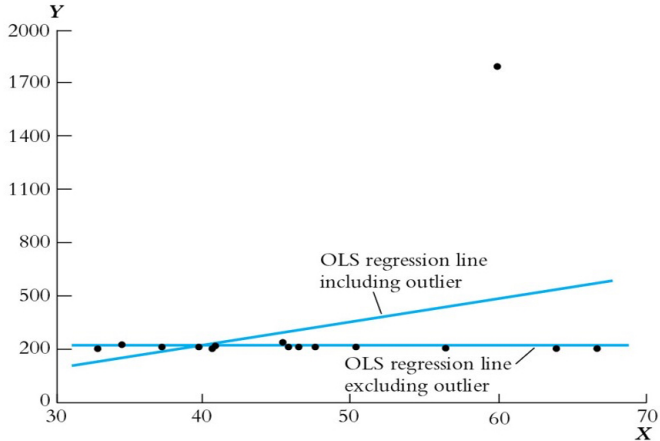
This arises automatically if the entity (individual, district) is sampled by simple random sampling:

- The entities are selected from the same population, so (X_i, Y_i) are identically distributed for all $i = 1, ..., n$.
- The entities are selected at random, so the values of (X, Y) for different entities are independently distributed.

The main place we will encounter non-i.i.d. sampling is when data are recorded over time for the same entity (panel data and time series data) – we will deal with that complication when we cover panel data.

Assumption (3): No large outliers

- If X and Y are bounded, then they have finite fourth moments. (Standardized test scores automatically satisfy this; family income, etc. satisfy this too.)
- The substance of this assumption is that a large outlier can strongly influence the results
- Why is it an outlier? Is it a typo? In practice, outliers are often data glitches (coding or recording problems). If so, they really shouldn't be in your data set!



OLS Sample Properties

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \\
&= \frac{\sum_{i=1}^{n}(X_i - \overline{X})Y_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \\
&= \frac{\sum_{i=1}^{n}(X_i - \overline{X})(\beta_0 + \beta_1 X_i + u_i)}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \\
&= \frac{\beta_0 \sum_{i=1}^{n}(X_i - \overline{X}) + \beta_1 \sum_{i=1}^{n}(X_i - \overline{X})X_i + \sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \\
&= \frac{\beta_1 \sum_{i=1}^{n}(X_i - \overline{X})(X_i - \overline{X}) + \sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \\
&= \beta_1 + \frac{\sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2}
\end{aligned}
$$

OLS Sample Properties — $\mathbb{E}[\hat{\beta}_1]$

$$
\begin{aligned}
\mathbb{E}[\hat{\beta}_1] &= \mathbb{E}[\mathbb{E}[\hat{\beta}_1 \mid X]] \\
&= \beta_1 + \mathbb{E}\left[\mathbb{E}\left[\frac{\sum_{i=1}^n (X_i - \overline{X})u_i}{\sum_{i=1}^n (X_i - \overline{X})^2}\middle| X\right]\right] \\
&= \beta_1 + \mathbb{E}\left[\frac{\mathbb{E}[\sum_{i=1}^n (X_i - \overline{X})u_i \mid X]}{\sum_{i=1}^n (X_i - \overline{X})^2}\right] \\
&= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^n (X_i - \overline{X})\mathbb{E}[u_i \mid X]}{\sum_{i=1}^n (X_i - \overline{X})^2}\right] \\
&= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^n (X_i - \overline{X})0}{\sum_{i=1}^n (X_i - \overline{X})^2}\right] \\
&= \beta_1
\end{aligned}
$$

OLS Finite Sample Properties — $var[\hat{\beta}_1]$

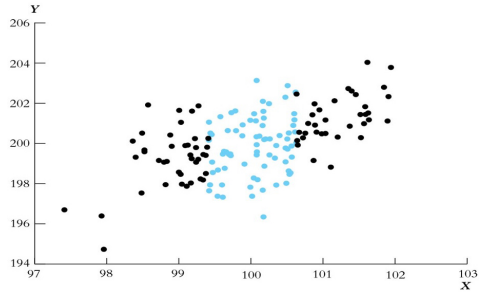
$$
\begin{aligned}
var(\hat{\beta}_1 \mid X) = var(\hat{\beta}_1 - \beta_1 \mid X) &= var\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \middle| X \right] \\
&= var\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})u_i}{TSS_X} \middle| X \right] \\
&= \frac{1}{TSS_X^2} var\left[\sum_{i=1}^{n}(X_i - \overline{X})u_i \middle| X \right] \\
&= \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2 var(u_i \mid X)}{TSS_X^2} \\
&= \frac{\sigma_u^2}{TSS_X}
\end{aligned}
$$

By the law of total variance (i.e., variance decomposition formula)

$$
var(\hat{\beta}_1) = \mathbb{E}(var(\hat{\beta}_1 \mid X)) + var(\mathbb{E}(\hat{\beta}_1 \mid X)) = \mathbb{E}(var(\hat{\beta}_1 \mid X)) = \frac{\sigma_u^2}{TSS_X}
$$

OLS Finite Sample Properties — $var[\hat{\beta}_1]$

- Here we made the assumption that u is homoskedastic (i.e., it has the same variance given any value of the explanatory variable): $var(u_i \mid X) = \sigma_u^2 \quad \forall \quad i$
- Note that a natural estimator of $var[\hat{\beta}_1]$, $\frac{\hat{\sigma}_u^2}{TSS_X}$ is $\frac{SSR}{(n-2)TSS_X}$.
- Larger error u variance increases $Var(\hat{\beta}_1)$ since it makes it harder to estimate β_1 accurately.



  - More variability in the independent variable (X) decreases the variance of $\hat{\beta}_1$ because it helps in estimating the relationship between Y and X.
  - Increasing sample size reduces the variance of $\hat{\beta}_1$ as it captures more variation in X.
  - Ideally, we would choose X to be as spread out as possible – this is sometimes possible with experimental data, but in practice, we often work with random samples

Large Sample Distribution of the OLS Estimator

- Other than its mean and variance, the exact distribution of $\hat{\beta}_1$ is complicated and depends on the distribution of $(X; u)$
- Because the $\mathbb{E}[\hat{\beta}_1] = \hat{\beta}_1$ and $var(\hat{\beta}_1) = 0$ as $n \to \infty$, then, by the Law of Large Numbers, $\hat{\beta}_1 \xrightarrow{p} \beta_1$ (i.e., $\hat{\beta}_1$ is a consistent estimator of β_1)
- When n is large, by the Central Limit Theorem, the sampling distribution of $\hat{\beta}_1$ is well approximated by a normal distribution $\mathcal{N}(\beta_1; \frac{\sigma_u^2}{TSS_X})$
- Said differently, $\frac{\hat{\beta}_1 - \beta_1}{\sqrt{var(\hat{\beta}_1)}} \sim \mathcal{N}(0; 1)$
- Note that this parallels the sampling distribution of $\overline{Y}$!
- We can use a similar reasoning with $\mathbb{E}[\hat{\beta}_0] = \hat{\beta}_0$ and that $var[\hat{\beta}_0] = \frac{\sigma_u^2 \sum_{i=1}^n X_i}{nTSS_X}$

$\Rightarrow$ We are now ready to tackle hypothesis testing with OLS!

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 4